

Atharva Girish Naik

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Summary

Backend Software Engineer with experience building scalable APIs, AI-powered systems, and cloud-native applications using Python and Node.js. Strong expertise in REST APIs, distributed systems, CI/CD, Docker, AWS, and database optimization. Passionate about building high-performance backend systems and production-ready DevOps workflows.

Technical Skills

Programming Languages: Python, JavaScript, TypeScript, SQL

Backend Frameworks: FastAPI, Node.js, Express

Cloud & DevOps: AWS (EC2, S3, IAM), Docker, GitHub Actions, CI/CD, Linux

Databases: PostgreSQL, MongoDB, ChromaDB

AI / ML: PyTorch, Scikit-learn, LangChain, RAG Pipelines

Testing & Performance: PyTest, Jest, Selenium, JMeter, API Testing, WebSocket Testing

Core Concepts: REST APIs, System Design, Microservices, OOP, Data Structures, Algorithms, Performance Optimization

Experience

Graduate Trainee | Tata Consultancy Services

May 2025 – Nov 2025

- Built an AI-powered VS Code extension automating bug detection, code fixes, and report generation, reducing manual review effort by 40%
- Designed and implemented a RAG-based semantic search pipeline using FastAPI, LangChain, and ChromaDB, improving retrieval accuracy by 30%
- Developed scalable REST APIs serving 10k+ requests/day and reduced average response time from 850ms to 480ms through query optimization and caching
- Optimized PostgreSQL queries using indexing and execution plan analysis, reducing database latency by 35%
- Implemented Docker-based CI/CD pipelines using GitHub Actions, decreasing deployment time by 50%
- Conducted API, WebSocket, and load testing using PyTest and JMeter, improving system reliability under concurrent traffic
- Collaborated in Agile sprints, contributing to production releases and peer code reviews

Backend Engineer | Perfect Engineering

Dec 2023 – May 2025

- Designed backend systems using Python and Node.js powering AI-driven workflows with modular microservice architecture
- Built ML pipelines using PyTorch and Scikit-learn, improving prediction accuracy by 25%
- Engineered REST APIs with PostgreSQL integration handling concurrent client requests efficiently
- Reduced infrastructure costs by containerizing applications using Docker and optimizing AWS EC2 resource utilization
- Implemented CI/CD pipelines via GitHub Actions, automating testing and reducing release cycle time by 40%
- Improved backend performance through caching strategies and database optimization techniques
- Contributed to distributed system design and AI agent workflow automation

Projects

AI News Aggregator (NewsAI)

Live Demo: newsai.atharvanaik.me

- Designed scalable backend architecture using FastAPI and PostgreSQL handling 5k+ daily requests
- Built AI-powered summarization pipeline with semantic retrieval (RAG-based architecture)
- Implemented caching and query optimization reducing API response time by 40%
- Containerized application using Docker and automated deployment with CI/CD

Smart VS Code Assistant

- Developed backend services for intelligent code analysis and automated bug detection
- Designed document retrieval APIs using RAG architecture with LangChain
- Reduced debugging time by 35% through automation workflows

Nakama AI Companion

- Built conversational backend services with secure and scalable REST APIs
- Optimized PostgreSQL queries improving performance and response latency

Education

Savitribai Phule Pune University

2024 – Present

Master of Computer Applications (MCA)

MMCC Pune

2021 – 2024

Bachelor of Computer Applications — CGPA: 9.49/10

Certifications

- Microsoft Azure DevOps Engineer Expert (AZ-400)
- Complete Machine Learning & Data Science Bootcamp
- DevOps + Web Development Cohort

Achievements

- Winner — Tech Setu Hackathon 2023
- Silver Medalist — C Debugging Competition
- 2nd Rank — Web Designing Competition
- Represented at Avishkar Zonal Level Research Project Competition